

Alcohol Intake, Age, BMI, and Gender as Key Determinants of Blood Pressure Variability: A Multilinear Regression Approach

How do physiological and lifestyle factors influence systolic blood pressure variability (BPV)?

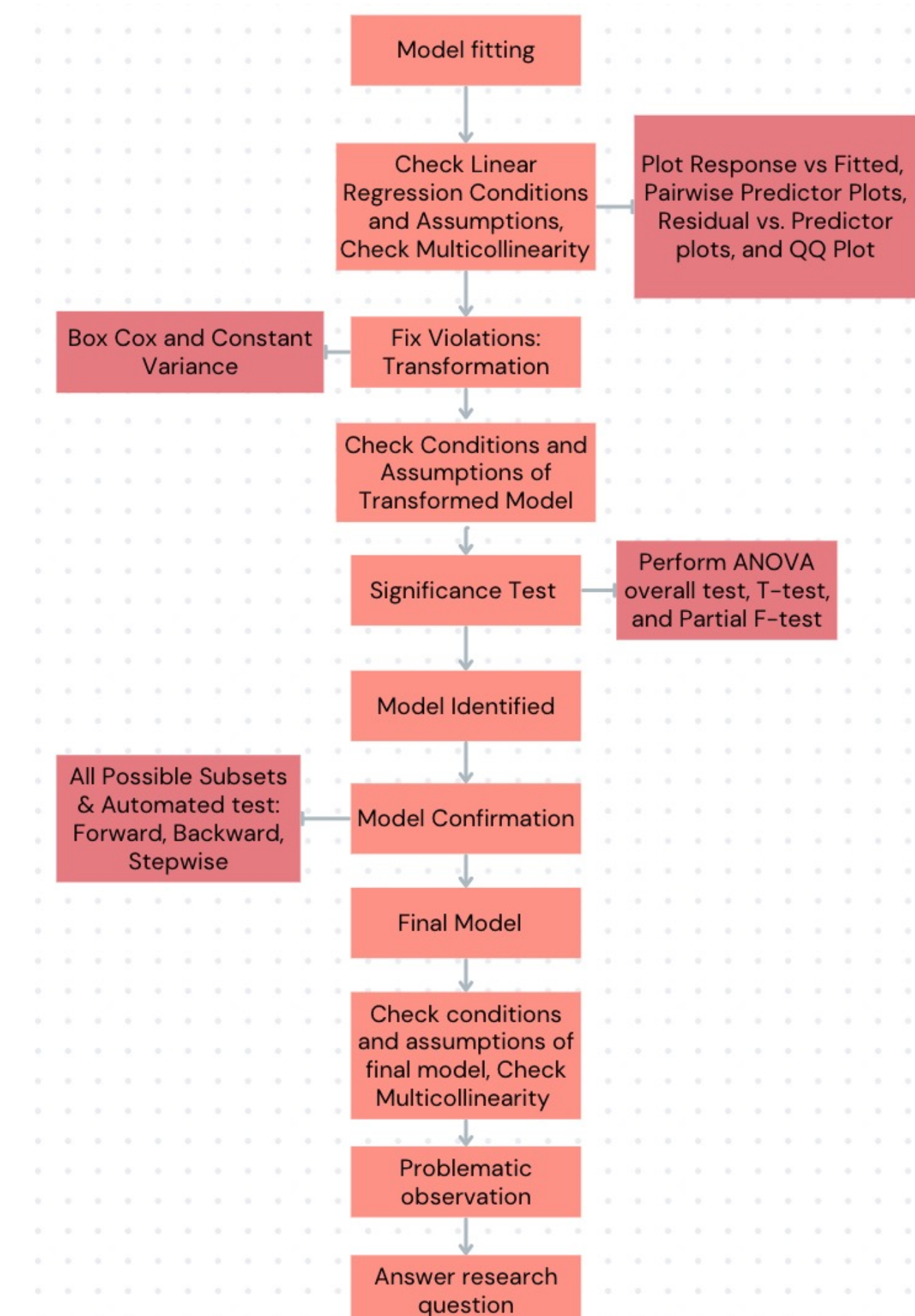
MOTIVATION

- High blood pressure, linked to heart disease and stroke, is a global health crisis driven by complex physiological and lifestyle factors (Martinez-Ríos et al., 2021)
- Gaining a deeper understanding of these factors is essential for developing effective prevention and enhancing quality of life. Our investigation enables a more comprehensive understanding of both physiological and lifestyle factors on BPV

DATA COLLECTION

- Source:** National Health and Nutrition Examination Survey (NHANES)
 - a publicly available and reliable dataset that undergoes an annual review and is governed by data protection laws
 - collected through interviews, physical examinations, and laboratory tests
- Sample size:** NHANES contains 10,000 entries of data, after data-cleaning, we ended up with 2925
- This data is an appropriate choice for the question because:**
 - Contains variables of interest: systolic blood pressure (BPSysAve), age, gender, BMI, alcohol intake, and physical activity.
 - Population diversity allows for generalizable insights into physiological and lifestyle factors affecting blood pressure variability.
 - Reliable and accurate measurement

METHODOLOGY

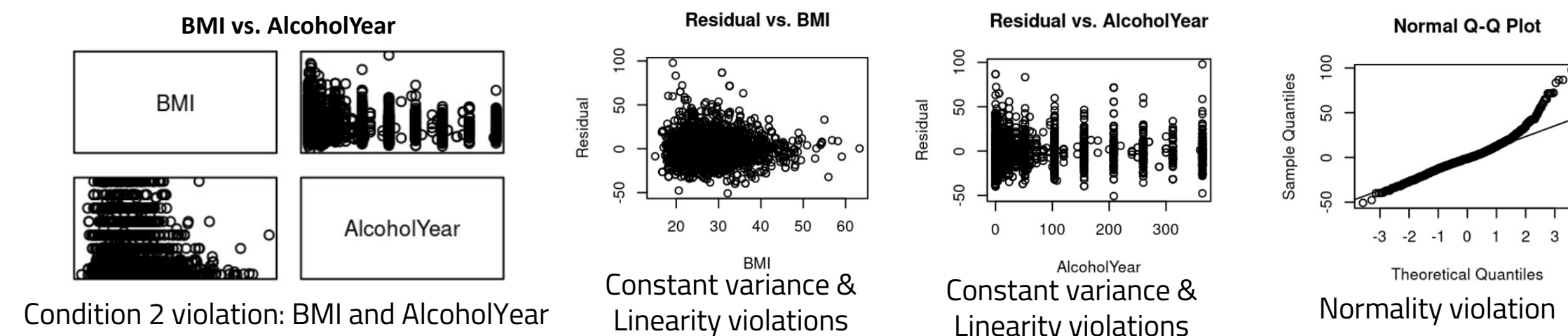


RESULTS

1. Initial model

$$BPSysAve \approx 88.588 + 5.165(Gender_{male}) + 0.365(Age) + 0.398(BMI) + 0.008(AlcoholYear) + 0.074(PhysActiveDays)$$

2. Condition & assumption violations



3. Multicollinearity of the initial model

- No severe multicollinearity in the initial model, VIF of all predictors are from 1.1-1.5

4. Transformed model built through Box-Cox Transformation

$$BPSysAve \approx 8.753 \times 10^3 - 3.856 \times 10^4 Gender - 2.898 \times 10^4 \sqrt{Age} + 1.021 \times 10^2 \frac{1}{\sqrt{BMI}} - 2.462 \times 10^5 \log(AlcoholYear + 1) - 1.774 \times 10^5 \sqrt{PhysActiveDays}$$

- Improved normality, constant variance, linearity, and condition 2 but still have a violation with uncorrelated errors

5. ANOVA overall significance test ($\alpha = 0.05$)

- p-value = $2.2e-16 < 0.05$
- This indicates that at least one predictor has a significant linear relationship with the response variable

6. T-test on coefficient from step 4 model

T-Test on Transformed Model Coefficients Table

Coefficient Name	P Values from T-test	Significance
Intercept	$9.088208e^{-237}$	Yes
Gender	$4.679556e^{-237}$	Yes
transAge*	$4.162989e^{-78}$	Yes
transBMI*	$8.899736e^{-27}$	Yes
transAlcoholYear*	0.01186504	Yes
transPhysActiveDays*	0.6347147	No

- transPhysActiveDays is a statistically insignificant predictor

*: $transAge = \sqrt{Age}$, $transBMI = \frac{1}{\sqrt{BMI}}$

$transAlcoholYear = \log(AlcoholYear + 1)$, $transPhysActiveDays = \sqrt{PhysActiveDays}$

7. Partial F-test

- Null Hypothesis (H_0): Removing transPhysActiveDays enables a simpler model but maintains comparable explanatory power as model in step 4.
- The F-value from the partial F-test (0.228) is greater 0.05, therefore we accept the simpler model that builds on the 4 predictors (Gender, transAge, transBMI and transAlcoholYear) and excludes transPhysActiveDays

Summary of Selected Model Table

Coefficient	Estimate	P Values
Intercept	8.772e-03	$< 2e-16$
Gender	-3.845e-04	$< 2e-16$
transAge*	-2.905e-05	$< 2e-16$
transBMI	1.019e-02	$< 2e-16$
transAlcoholYear	-2.475e-05	0.0114

8. All Possible Subsets & Automated Selection

Best Models from All Possible Subsets Method Table

Number of Predictors	Predictors of Best Model with Constraint	Adjusted R Squared
1	transAge	0.1118534
2	Gender transAge	0.1502323
3	Gender transAge transBMI	0.1811116
4	Gender transAge transBMI transAlcoholYear	0.1826249
5	Gender transAge transBMI transAlcoholYear transPhysActiveDays	0.1824081

- All possible subsets method identifies the model with the largest adjusted R^2 to be the same model selected by the significance test, the model with 4 predictors.
- All automated selection methods (forward, backward, stepwise) select the same model.

Conclusions

Final Model

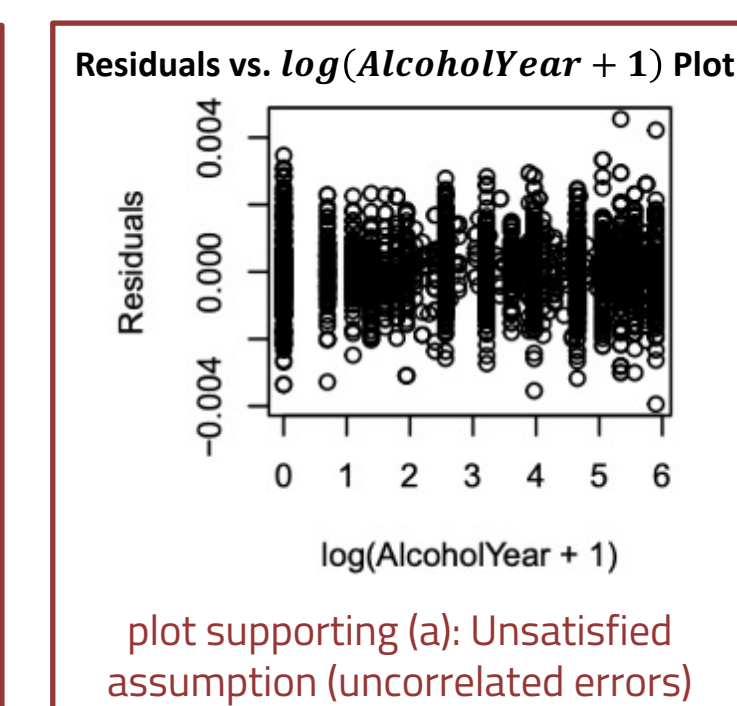
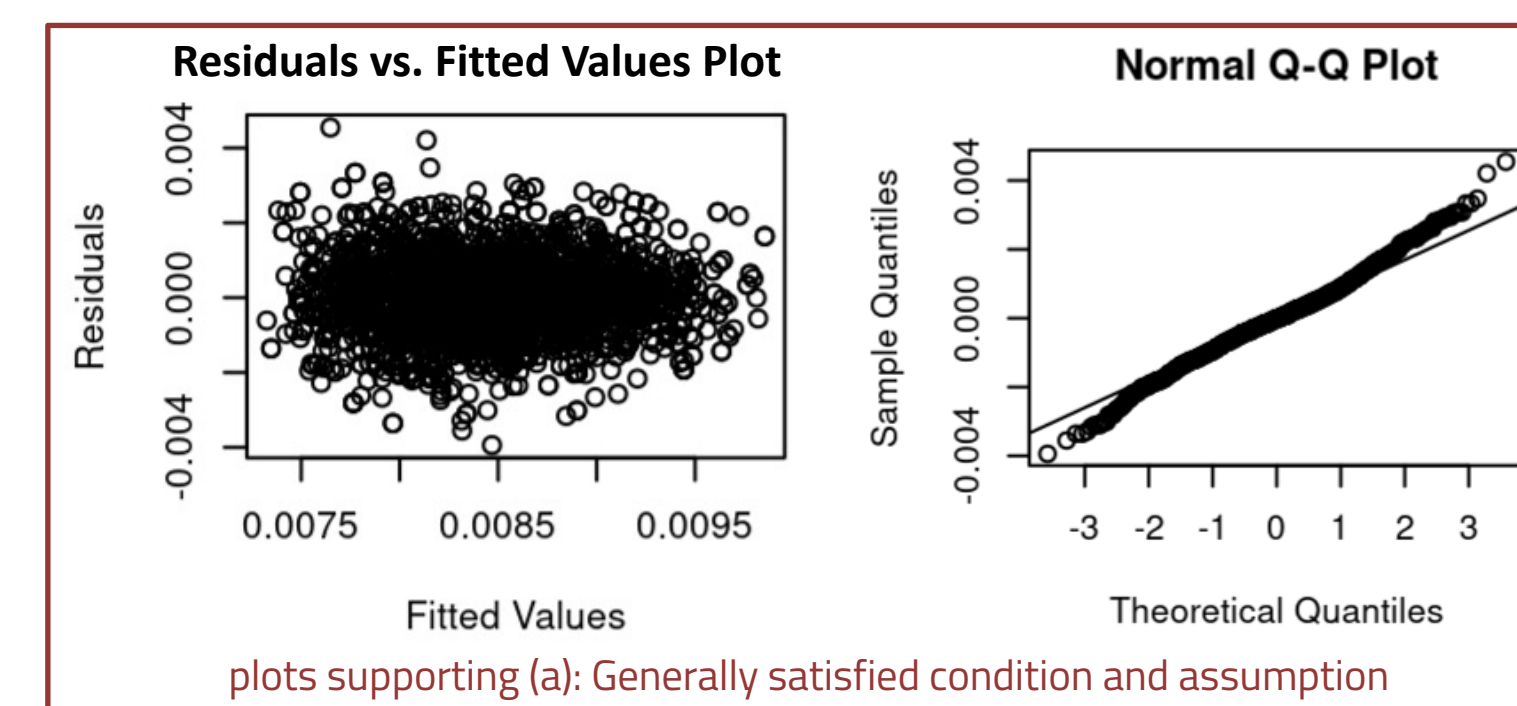
$$BPSysAve \approx 8.728 \times 10^3 - 3.845 \times 10^4 Gender - 2.905 \times 10^4 \sqrt{Age} + 1.019 \times 10^2 \frac{1}{\sqrt{BMI}} - 2.475 \times 10^5 \log(AlcoholYear + 1)$$

Key Findings from Our Final Model:

- All predictors—age, gender, BMI, and annual alcohol consumption frequency—have a significant impact on expected BPV, contributing to one's chance.
- When one is older, has bigger BMI, and consumes alcohol more frequently, the expected BPV is higher. Moreover, males with the same measurements for the other predictors have a higher BPV compared to females.

Model Validation and Assumption Check:

- All linear regression conditions and assumptions are generally satisfied for the final selected model
- No severe multicollinearity in the final model. VIF of all predictors are approximately 1.
- No leverage points, no influential points on all fitted values, no influential points on its own values, 23 influential points on all estimated coefficients, and three outlier points



VIF table for (b)

Coefficient	VIF
Gender	1.024222
transAge	1.011593
transBMI	1.037512
transAlcohol	1.038079

LIMITATIONS

Accuracy of our model for answering the research question:

- Final model is not entirely accurate, which can be seen through a low adjusted R-squared value (0.1826). This suggests it explains only a small portion of the variability in BPV.
- Multiple reasons can contribute to this lack of accuracy:
 - Uncorrelated Error Violation:** We can see serious clustering for uncorrelated errors. This violation will make our standard error estimates unreliable, potentially leading to over- or under-estimation of predictor variances.
 - Too few predictors:** The complexity of hypertension and BPV likely involves additional influencing factors not included in the model. The initial literature search may not have identified all relevant variables.

Suggestion:

- Read more literature to gather more potential predictors for the model. This may help increase the adjusted R-squared, improving the proportion of variation that our model can explain on BPV.

REFERENCES

- Martinez-Ríos, E., Montesinos, L., Alfaro-Ponce, M., & Pecchia, L. (2021). A review of machine learning in hypertension detection and blood pressure estimation based on clinical and physiological data. *Biomedical Signal Processing and Control*, 68, 102813. <https://doi.org/10.1016/j.bspc.2021.102813>
- Centers for Disease Control and Prevention. (2022, February 16). *National Health and Nutrition Examination Survey (NHANES): Methodology*. National Center for Health Statistics. <https://www.cdc.gov/nchs/hus/sources-definitions/nhanes.htm>